

Probability Assignment 1

Exercise 2.1

Q2.6) a) The first column consists of qualitative data. This column provides the names of the states in which disciplinary actions were recorded.

b) The second column consists of quantitative, discrete data. This column provides the number of serious disciplinary actions taken against doctors in each state. Since these are counts of actions, they can take only whole number values.

c) The third column consists of quantitative, continuous data. This column provides the rate of serious disciplinary actions per 1000 doctors. Since this is a ratio that can include decimal values, it's continuous.

Q2.7a) The second column consists of quantitative, discrete data. This column provides ranks of the highest recorded temperature for each city.

b) The third column consists of quantitative, continuous data. This column provides the highest recorded temperatures for the city in degrees Fahrenheit.

c) The information that Phoenix is in Arizona is qualitative data since the state in which a city is located is non numerical.

Q2.8a) The first column consists of quantitative, discrete data. This column provides the ranks of the deceased celebrities based on their earnings. Since rank values are whole numbers, they are discrete.

b) The third column consists of quantitative, continuous data. This column provides the earnings in million of dollars. Since earnings can take decimal values, the data is continuous.

Q2.9a) The first column consists of quantitative, discrete data. This column provides the rank of each country based on the number of Wi-fi locations. Since rank value are whole numbers they are discrete.

b) The second column consists of qualitative data. This column provides the names of the countries.

c) The third column consists of quantitative discrete data. This column provides the number of Wi-fi locations in each country. Since these are counts, they can take only whole-number values.

Q2.10a) The first column consists of qualitative data. This column provides the type of product (CD, Cassette, DVD etc)

b) The second column consists of quantitative, continuous data. This column provides the number of units shipped in million. Since the values include decimals, the data is continuous.

c) The third column consists of quantitative, continuous data. This column

provides the retail dollar value in millions. Since dollar value can include decimals, the data is continuous.

Exercise 2.2

Q2.19a) Frequency Distribution Table and Relative Frequency

Champion	Tally Marks	Frequency	Relative Frequency
Iowa		13	$\frac{13}{25} = 0.52$
Oklahoma St		7	$\frac{7}{25} = 0.28$
Minnesota		3	$\frac{3}{25} = 0.12$
Iowa St		1	$\frac{1}{25} = 0.04$
Arizona St		1	$\frac{1}{25} = 0.04$

$$\Sigma F = 25$$

$$\Sigma R.F = 1$$

c) Pie Chart

$$\text{Angle} = R.F \times 360$$

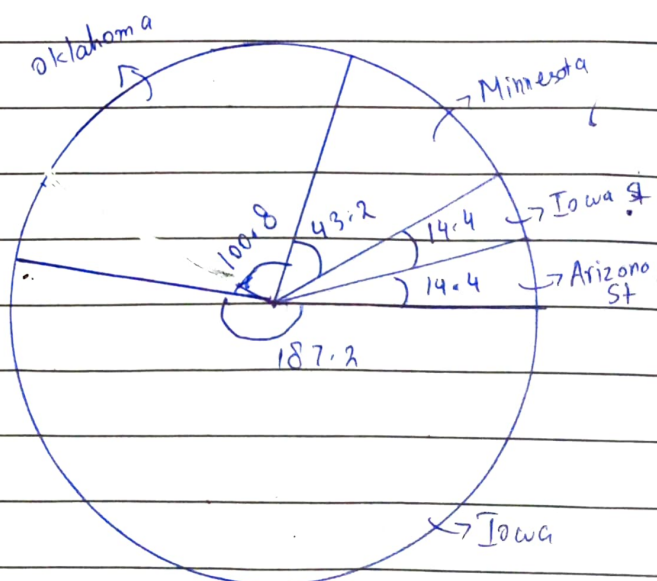
$$\text{so Iowa} = 0.52 \times 360 = 187.2$$

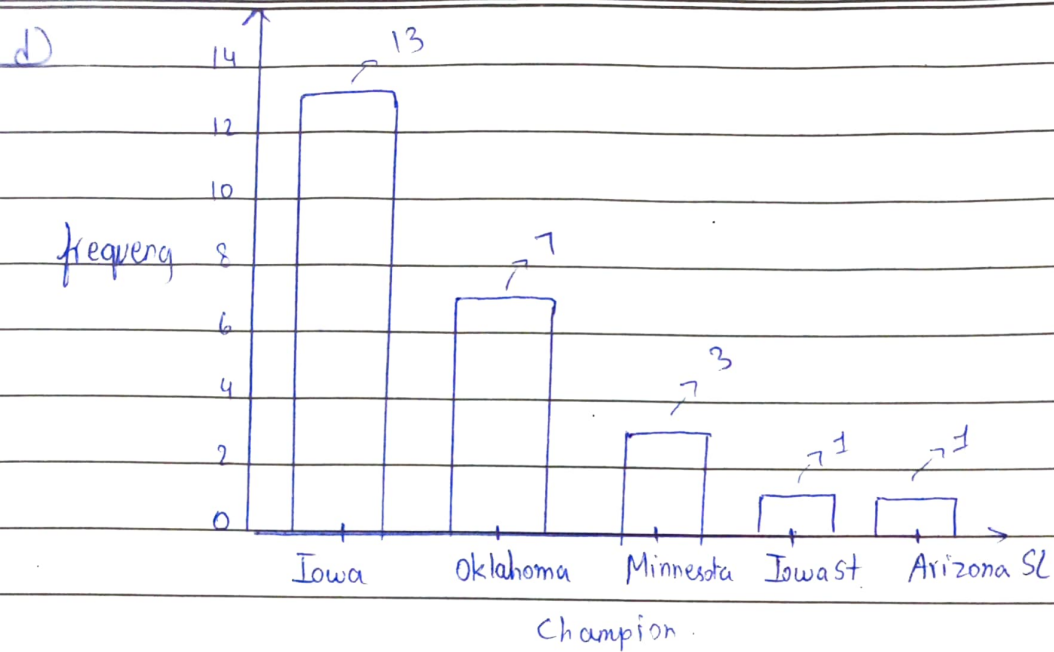
$$\text{Oklahoma} = 0.28 \times 360 = 100.8$$

$$\text{Minnesota} = 0.12 \times 360 = 43.2$$

$$\text{Iowa St} = 0.04 \times 360 = 14.4$$

$$\text{Arizona St} = 0.04 \times 360 = 14.4$$



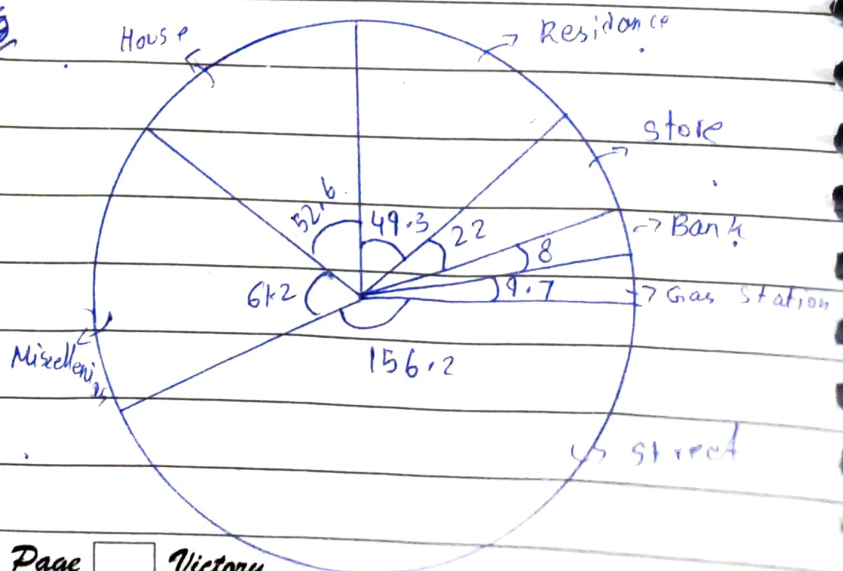


Robbery Type	Frequency	Relative Frequency (3d.p)
Street	179 296	$\frac{179296}{413403} = 0.434$
Commercial House	60 493	$\frac{60493}{413403} = 0.146$
Gas station	11 362	$\frac{11362}{413403} = 0.027$
Convenience store	25 774	$\frac{25774}{413403} = 0.062$
Residence	56 641	$\frac{56641}{413403} = 0.137$
Bank	9 504	$\frac{9504}{413403} = 0.023$
Miscellaneous	70 333	$\frac{70333}{413403} = 0.170$

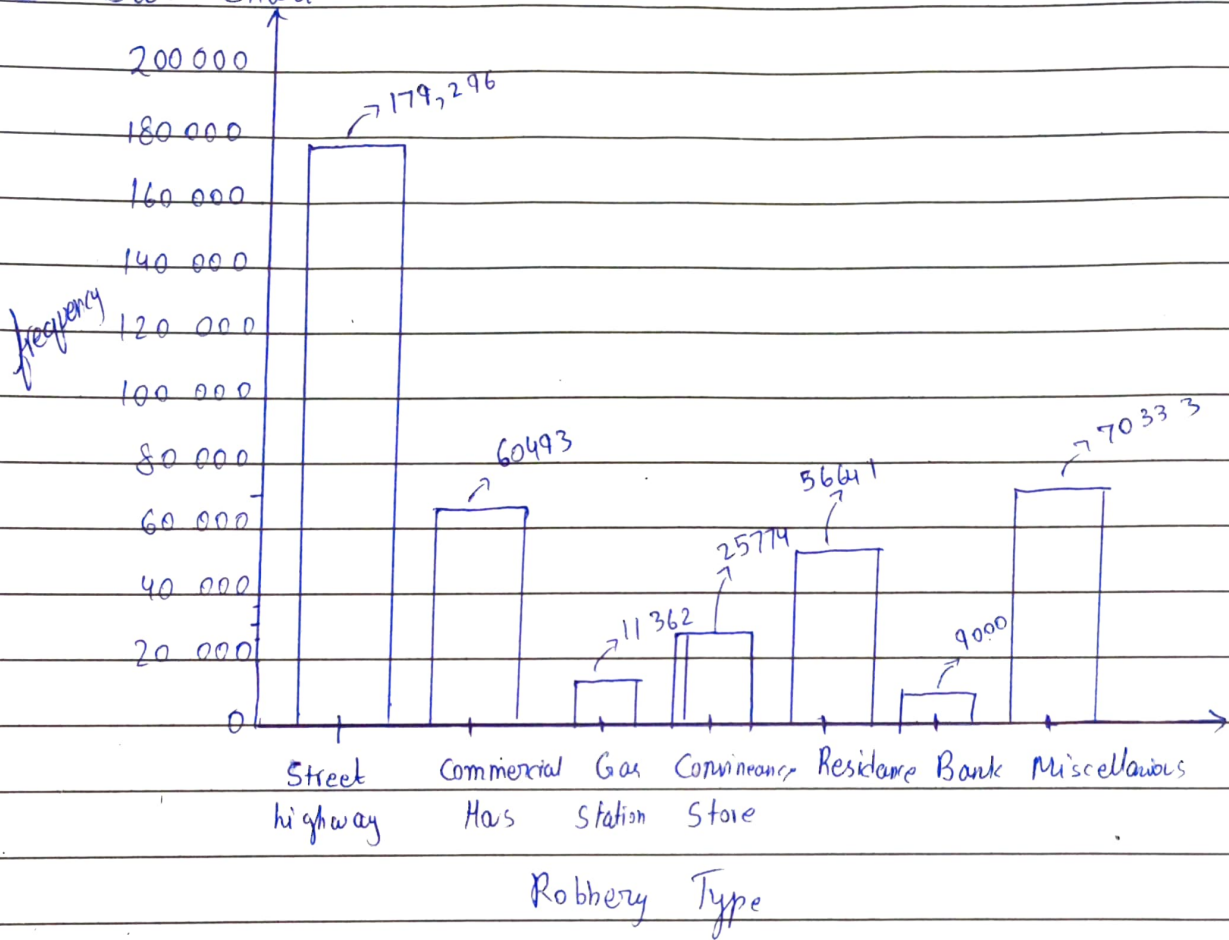
$$\Sigma f = 413403$$

b) Pie chart

- street = 156.24°
- ✓ House = 52.6°
- ✓ Gas station = 9.7°
- Store = 22.32
- ✓ Residence = 49.32
- ✓ Bank = 8.3
- ✓ Miscellaneous = 61.2



d) Bar Chart



Exercise 2.3

Q2.56) Data (n) = 50 Min = 45 Max = 155~~Range = 110 class width = $\frac{110}{11} = 10$~~

class ranges is 40-49

class width = 10

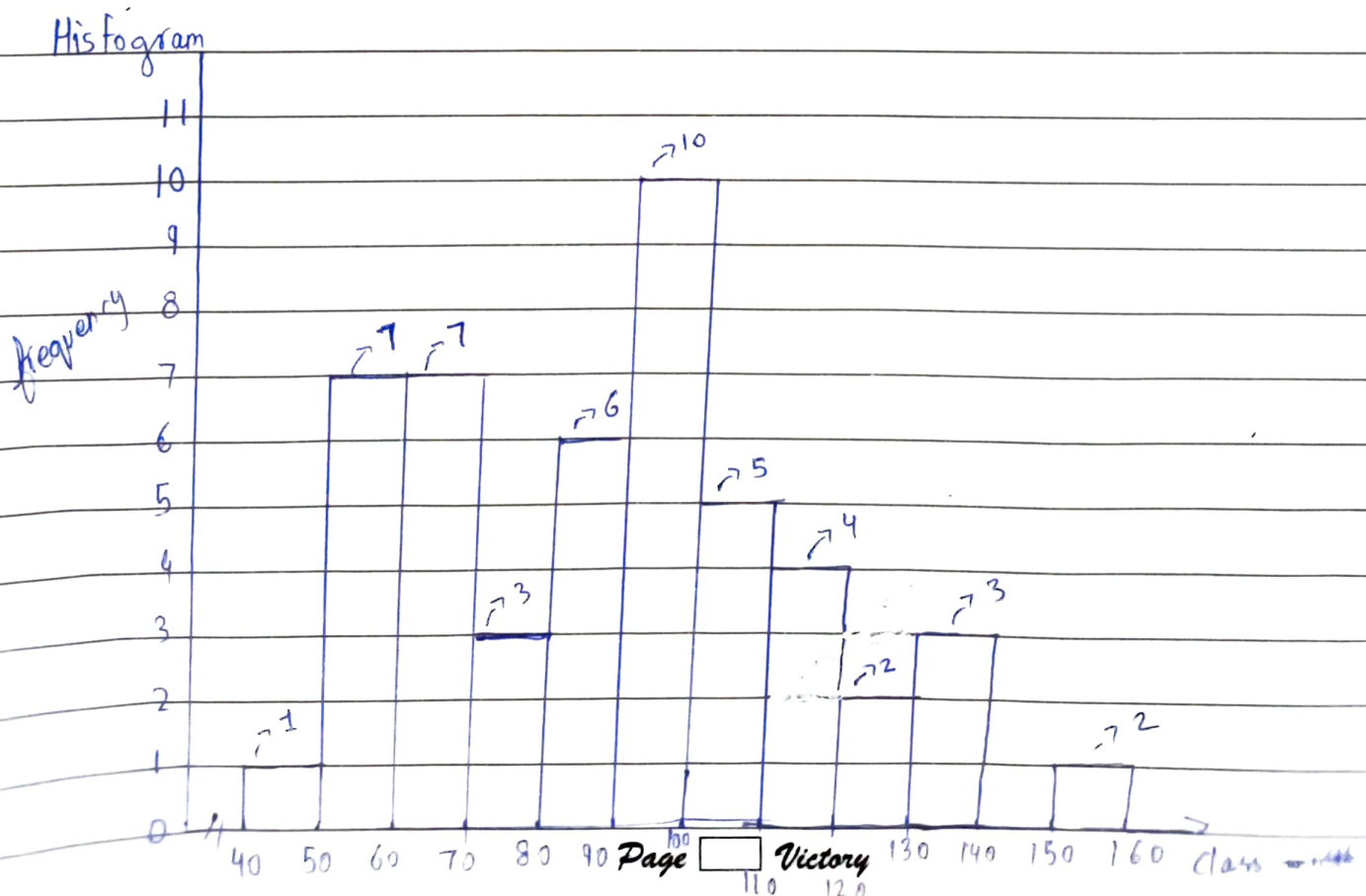
$$\frac{160-40}{10} = 12 \Rightarrow \text{classes to be made}$$

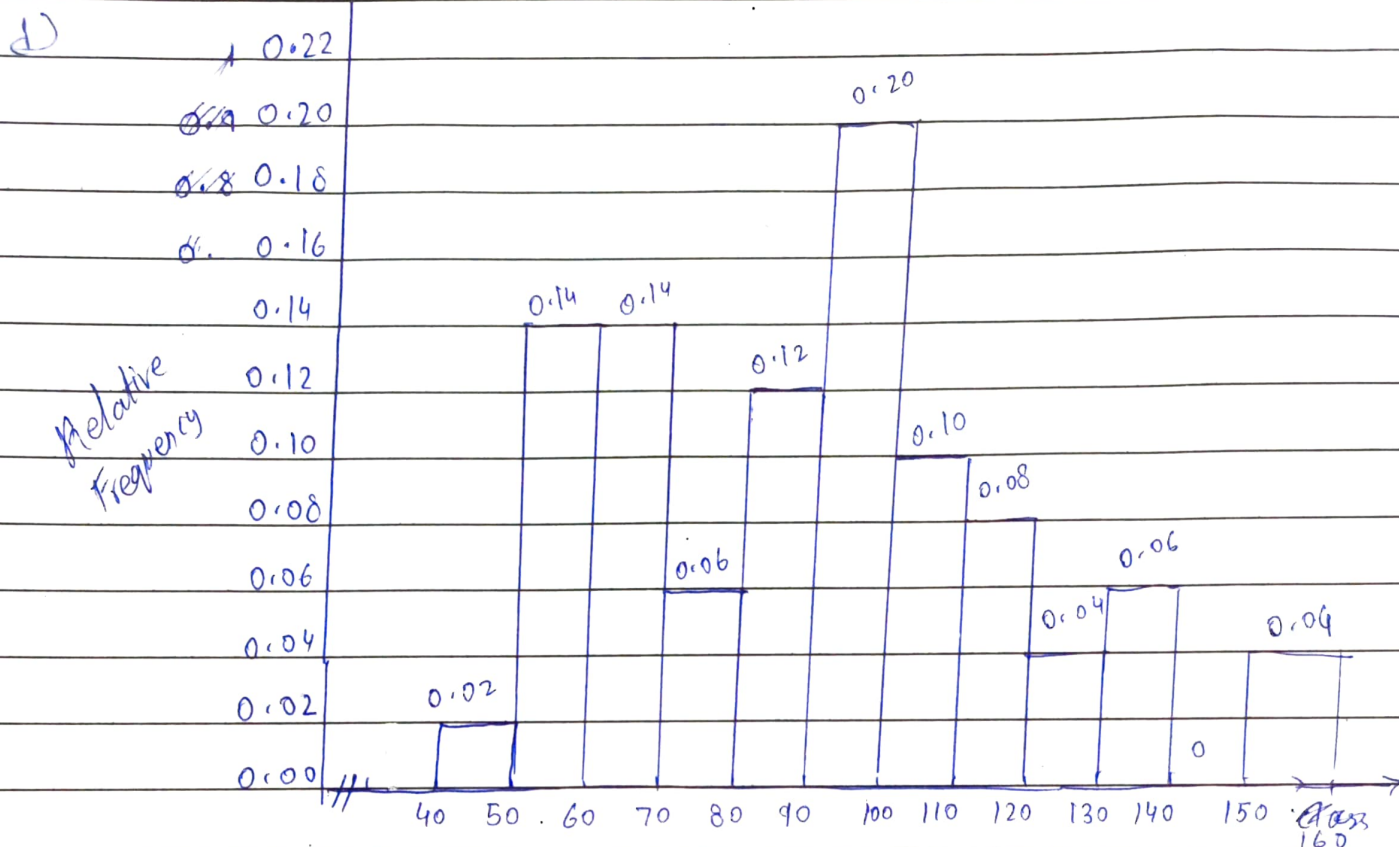
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class	Frequency	Relative Frequency
40 - 49	1	$\frac{1}{50} = 0.02$
50 - 59	7	$\frac{7}{50} = 0.14$
60 - 69	7	$\frac{7}{50} = 0.14$
70 - 79	3	$\frac{3}{50} = 0.06$
80 - 89	6	$\frac{6}{50} = 0.12$
90 - 99	10	$\frac{10}{50} = 0.2$
100 - 109	5	$\frac{5}{50} = 0.1$
110 - 119	4	$\frac{4}{50} = 0.08$
120 - 129	2	$\frac{2}{50} = 0.04$
130 - 139	3	$\frac{3}{50} = 0.06$
140 - 149	0	$\frac{0}{50} = 0$
150 - 159	2	$\frac{2}{50} = 0.04$

$$\Sigma f = 50$$

$$\Sigma R.F = 1$$

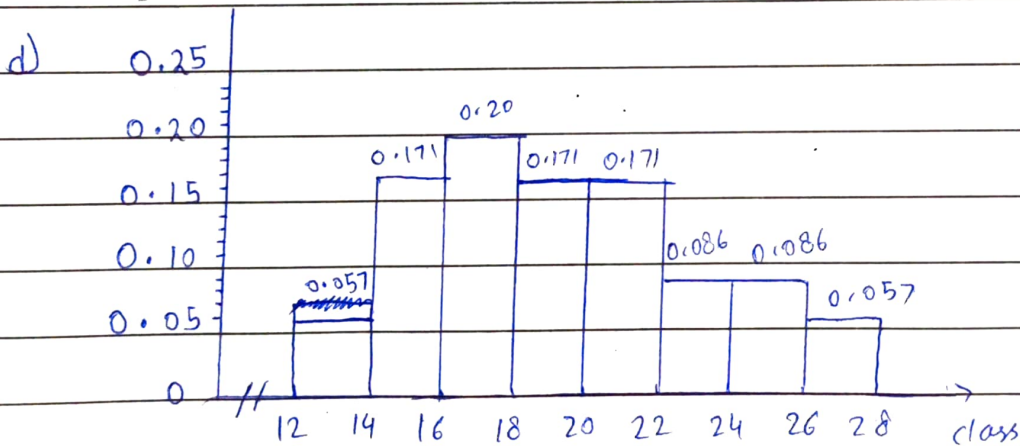
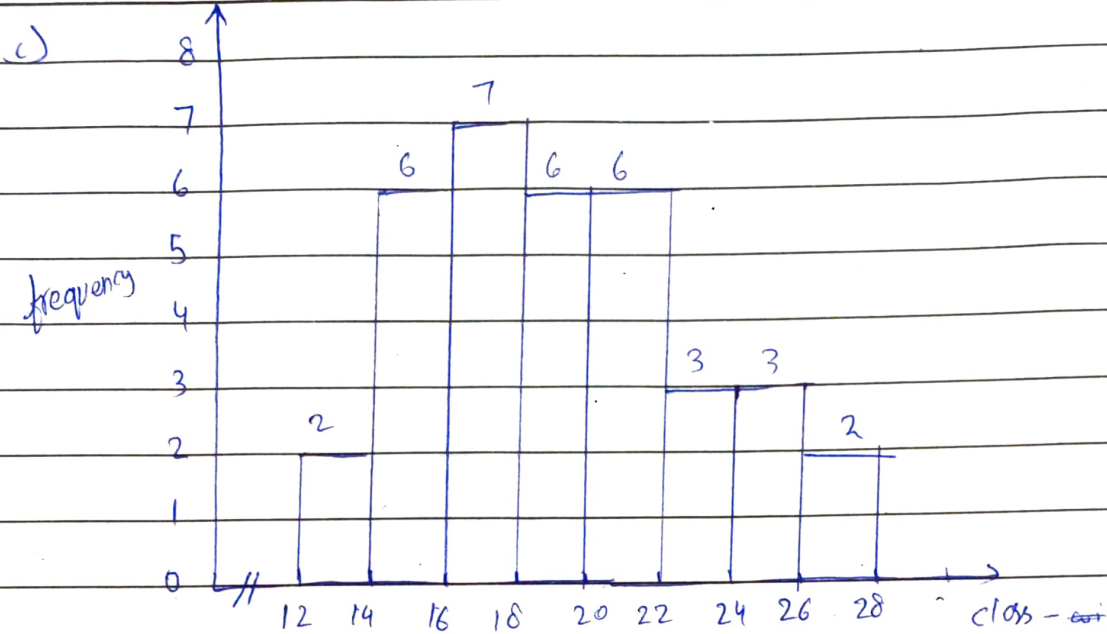




Q2.62)

class	Frequency	Relative Frequency (3.dp)
12 - under 14	2	$\frac{2}{35} = 0.057$
14 - under 16	6	$\frac{6}{35} = 0.171$
16 - under 18	7	$\frac{7}{35} = 0.200$
18 - under 20	6	0.171
20 - under 22	6	0.171
22 - under 24	3	$\frac{3}{35} = 0.085$
24 - under 26	3	0.085
26 - under 28	2	$\frac{2}{35} = 0.057$

$$\Sigma f = 35$$

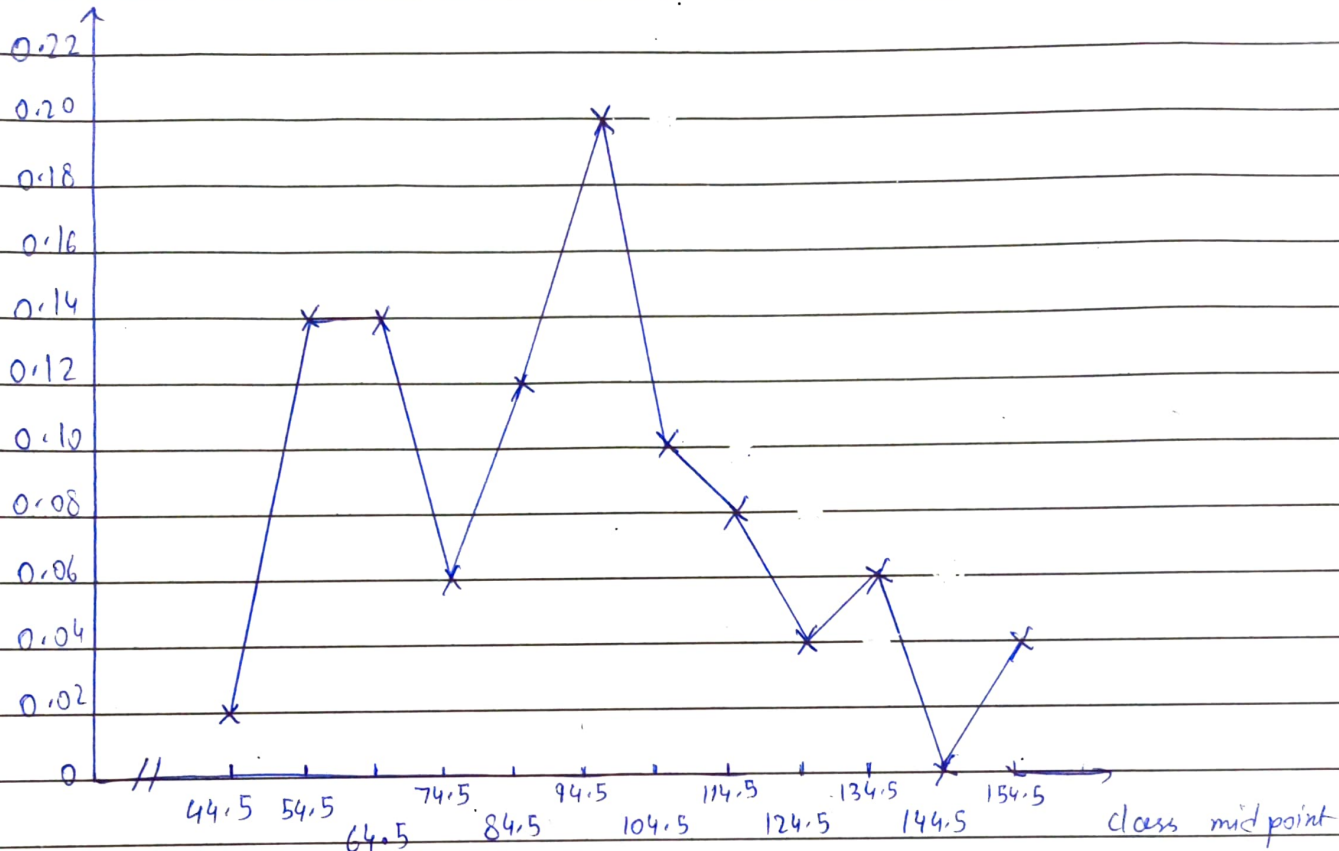


Q 2.85)

class	Mid-point	R.F	Class	Mid-point	R.F
44.5		0.02	124.5		0.04
54.5		0.14	134.5		0.06
64.5		0.14	144.5		0
74.5		0.06	154.5		0.04
84.5		0.12			
94.5		0.2			
104.5		0.1			
114.5		0.08			

Relative frequency

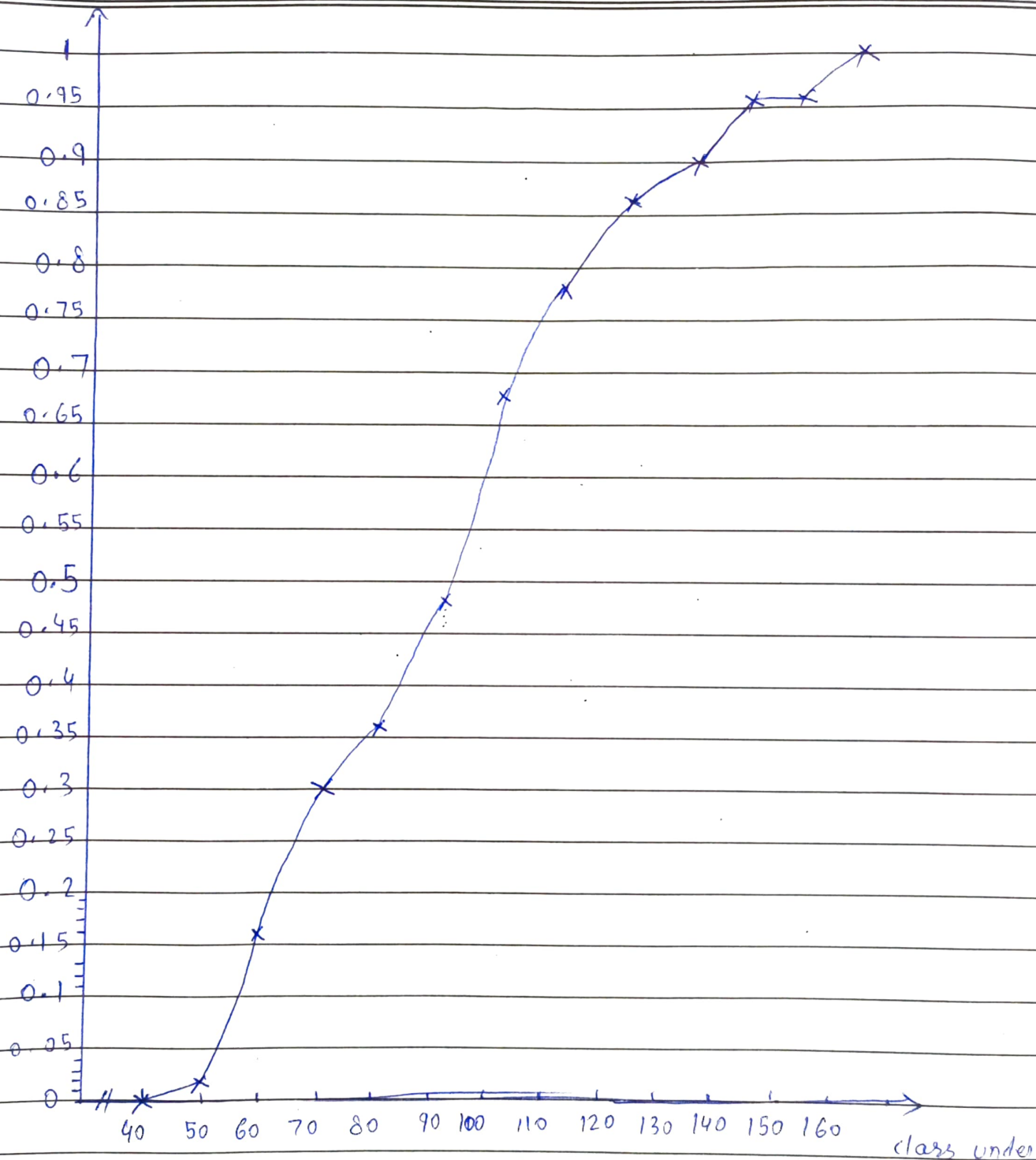
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Q2.88a)	less than	cummulative Frequency	Cummulative Relative Frequency
	40	0	0
	50	$0+1 = 1$	0.02
	60	$1+7 = 8$	0.16
	70	$8+7 = 15$	0.30
	80	$15+3 = 18$	0.36
	90	$18+6 = 24$	0.48
	100	$24+10 = 34$	0.68
	110	$34+5 = 39$	0.78
	120	$39+4 = 43$	0.86
	130	$43+2 = 45$	0.90
	140	$45+3 = 48$	0.96
	150	$48+0 = 48$	0.96
	160	$48+2 = 50$	1

cumulative
Relative frequency

Date: _____



Ogive Graph

Exercise 3.3

Q3.123) Data in Ascending order

1 1 3 3 4 4 5 6 6 7 7 9 9 10 12 12 13
15 18 23 55

Total items: 21

$$a) \text{ Lower Quartile } \Rightarrow Q_1 = \frac{i(n+1)}{4}$$

$$= \frac{1(21+1)}{4}$$

$$= \frac{22}{4}$$

$$= 5.5^{\text{th}} \text{ position}$$

$$5^{\text{th}} \text{ value} + 0.5 \cdot (6^{\text{th}} \text{ val} - 5^{\text{th}} \text{ val})$$

$$= 4 + 0.5(4 - 4)$$

$$\therefore = 4 \Rightarrow Q_1 \text{ value}$$

$$\text{Median (Mid Quartile)} \Rightarrow Q_2 = \frac{i(n+1)}{4}$$

$$= \frac{2(21+1)}{4}$$

$$= \frac{22}{2}$$

$$= 11^{\text{th}} \text{ position}$$

$$\therefore Q_2 = 7$$

$$\text{Upper Quartile } \Rightarrow Q_3 = \frac{i(n+1)}{4} = \frac{3(21+1)}{4} = \frac{3(22)}{4} = 16.5^{\text{th}} \text{ pos}$$

$$16^{\text{th}} \text{ pos} + 0.5(17^{\text{th}} \text{ pos} - 16^{\text{th}} \text{ pos})$$

$$= 12 + 0.5(13 - 12) = 12.5$$

Quantiles

$Q_1 = 4$ $\{ 25\% \text{ of patients stay } \leq 4 \text{ days}$

$Q_2 = 7$ $\{ 50\% \text{ of patients stay } \leq 7 \text{ days}$

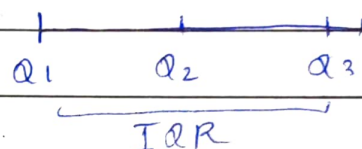
$Q_3 = 12.5$ $\{ 75\% \text{ of patients stay } \leq 12.5 \text{ days}$

b) Inter Quartile Range (IQR)

$$IQR = Q_3 - Q_1$$

$$= 12.5 - 4$$

$$= 8.5$$



Interpretation: The middle 50% of hospital stays range over 8.5 days

c) Five Number Summary

Minimum = 1

First Quartile (Q_1) = 4

Median (Q_2) = 7

Third Quartile (Q_3) = 12.5

Maximum = 55

Interpretation: Most hospital stays are concentrated between 1 and 23 days, with an unusually long stay at 55 days

d) Outlier

$$\text{Lower-limit} = Q_1 - 1.5 (IQR)$$

$$= 4 - 1.5 (8.5)$$

$$= -8.75$$

$$\text{Upper-limit} = Q_3 + 1.5 (IQR)$$

$$= 12.5 + 1.5 (8.5)$$

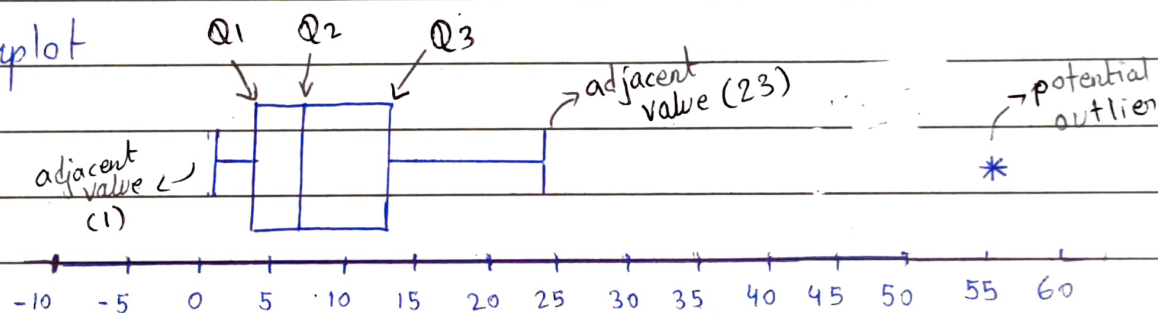
$$= 25.25$$

Outliers value are any value < -8.75 and any value > 25.25

So from data we have none value that is less than -8.75 but from data we have one value that is greater than 25.25 which is 55

Potential Outlier = 55 days

e) Boxplot



Adjacent values are 1 and 23 as 55 is a potential outlier

$$Q_1 = 4$$

$$Q_2 = 7$$

$$Q_3 = 12.5$$

Interpretation :- Most hospital stays (middle 50%) range from 4 to 12.5 days

Median stay is 7 days and is slightly Right skewed (because maximum is large)

One patient had a very long stay (55 days) which is potential outlier.

Q) 3.127

Arrange data in ascending order

21 70 125 195 389 649 656 664 682 1006 1300
 1403 1433 1800 1982 2205 2515 3027 3634 4200
 5299 5947 7886 8543 9310 11189 17341

Total items = 27

$$a) \text{ Lower Quartile } (Q_1) \Rightarrow Q_1 = \frac{i(n+1)}{4} = \frac{1(27+1)}{4} = \frac{28}{4} = 7^{\text{th}} \text{ position}$$

$$Q_1 = 656$$

$$\text{Median } (Q_2) \Rightarrow Q_2 = \frac{i(n+1)}{4} = \frac{2(27+1)}{4} = \frac{28}{2} = 14^{\text{th}} \text{ position}$$

$$Q_2 = 1800$$

$$\text{Upper Quartile } (Q_3) \Rightarrow Q_3 = \frac{i(n+1)}{4} = \frac{3(27+1)}{4} = 3(7) = 21^{\text{th}} \text{ position}$$

$$Q_3 = 5299$$

Interpretation

$Q_1 = 656$ { 25% of companies spent $\leq \$656,000$
 $Q_2 = 1800$ { 50% of companies spent $\leq \$1800,000$
 $Q_3 = 5299$ { 75% of companies spent $\leq \$5299,000$

b) Inter Quartile Range

$$IQR = Q_3 - Q_1$$

$$= 5299 - 656$$

$$= 4643$$

Interpretation:

The middle 50% of companies spent \$ 4643 000 in capital spending.

c) Five Number Summary

$$\text{Minimum} = 21$$

$$\text{First Quartile } (Q_1) = 656$$

$$\text{Median } (Q_2) = 1800$$

$$\text{Third Quartile } (Q_3) = 5299$$

$$\text{Maximum} = 17341$$

Interpretation :- Most companies spend between \$21,000 and \$11,189,000 with a few extremely high spenders. (17,341,000)

d) Outliers

$$\text{Lower Limit} = Q_1 - 1.5(IQR)$$

$$= 656 - 1.5(4643)$$

$$= -6308.5$$

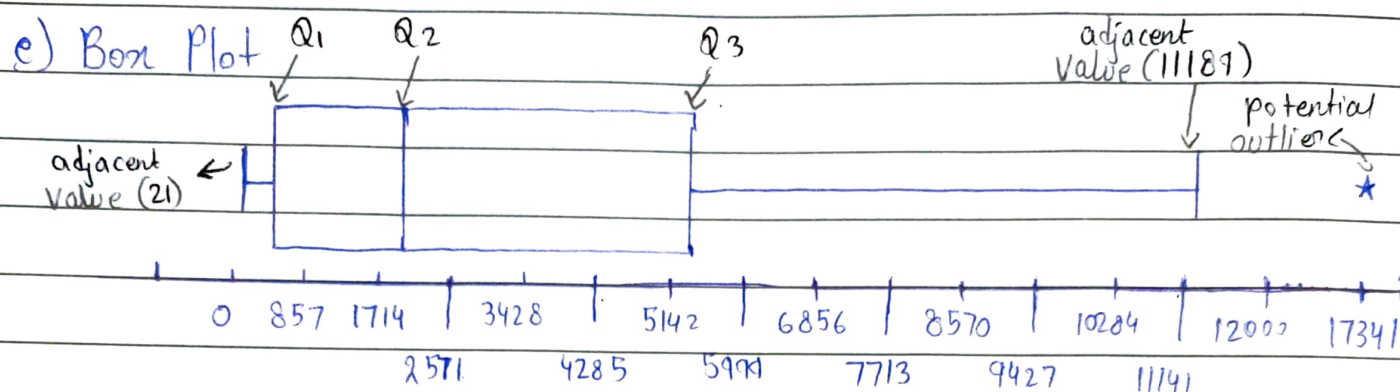
$$\text{Upper Limit} = Q_3 + 1.5(IQR)$$

$$= 5299 + 1.5(4643)$$

$$= 12263.5$$

Outliers value are any value < -6308.5 and
any value > 12263.5

Potential Outlier = 17341



Adjacent values are 21 and 11,189 as 17341 is an outlier

$$Q_1 = \cancel{21} 656$$

$$Q_2 = 1800$$

$$Q_3 = \cancel{5294} 5294$$

Interpretation: Most companies spending is concentrated between \$656,000 and \$5.3M

Median spending is \$1.8M and is skewed right

One company is an extreme outlier (\$17341 k)